Self-Check

Lecture 0

Prerequisites: A Refresher

This sheet begins with a short diagnostic of the four computational skills the early lectures rely on. Take only the seven problems listed below *cold*, without notes, and stop after 45 minutes. Then check your work against the answers in the student edition of this sheet (or with your tutor) and use the routing guide at the end: repair only the weak strand and complete its recheck. The remaining algebra problems form a targeted bank, not a second cold test. Throughout, $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$.

Scored algebra diagnostic (40–45 minutes). Attempt problems **0.1, 0.2, 0.9, 0.15, 0.22, 0.23, and 0.30**. Award one point only for a correct result supported by enough working to identify the method. These seven checks cover complex arithmetic and polar form, compatible matrix operations, determinants, row reduction, null-space bases, and consistency-first classification.

- **Ready:** 7/7. A single miss is not "ready": each check stands for a distinct skill, so missing 0.2 means polar form and missing 0.23 means null-space bases, whatever the total. Take the route for whichever check you missed.
- **Rebuild first:** gaps in two or more of the four strands, or 0–3 points. Work through each weak route with notes open, then retake the seven diagnostic checks on a later day.
- **Targeted repair:** anything else—that is, 4–6 points with at most one weak strand. Use the matching refresher section and bank route, then complete the named recheck before Lecture 1.

Separate analysis-readiness route. Problems 0.32–0.36 are a non-scored, 15–20 minute routing check for differentiation, integration by parts, elementary linear ODEs, series, and Fourier coefficients. They are not part of the seven-point algebra score and are not assumed in Lecture 0 itself. Use them before the later application weeks; this sheet is not a replacement calculus or analysis course.

Complex arithmetic

Exercise 0.1 (diagnostic) Let $z = 3 + 2i$ and $w = 1 + i$. Compute $z + w$, zw, z/w, the conjugate $\overline{z}$, and the modulus $|z|$.

Answer. $z + w = 4 + 3i$; $\quad zw = 1 + 5i$; $\quad z/w = \frac{5}{2} - \frac{1}{2}i$; $\quad \overline{z} = 3 - 2i$; $\quad |z| = \sqrt{13}$.

Exercise 0.2 (diagnostic) Write $1 - i$ in polar form and use it to compute $(1 - i)^8$.

Answer. $1 - i = \sqrt{2}\, e^{-i\pi/4}$, so $(1 - i)^8 = 16$.

Exercise 0.3 (computational) Find all complex numbers z with $z^2 = i$.

Hint. Write i in polar form and take square roots: halve the argument, and remember a quadratic has two roots.

Answer. $z = \pm\dfrac{1 + i}{\sqrt{2}} = \pm\, e^{i\pi/4}$.

Exercise 0.4 (computational) Let $z = 3 + i$ and $w = 1 + 2i$. Compute $z + w$, zw, z/w, the conjugate $\overline{z}$, and the modulus $|z|$.

Answer. $z+w=4+3i$; $zw=1+7i$; $z/w=1-i$; $\overline{z}=3-i$; $|z|=\sqrt{10}$.

Exercise 0.5 (computational) Write $1+i$ in polar form and use it to compute $(1+i)^6$.
Answer. $1+i=\sqrt{2}\,e^{i\pi/4}$, so $(1+i)^6=-8i$.

Exercise 0.6 (computational) Write $\sqrt{3}+i$ in polar form and use it to compute $(\sqrt{3}+i)^4$, giving the answer in Cartesian form.
Answer. $\sqrt{3}+i=2\,e^{i\pi/6}$, so $(\sqrt{3}+i)^4=-8+8\sqrt{3}\,i$.

Exercise 0.7 (bridge) For complex numbers $z=a+bi$ and $w=c+di$, prove that $\overline{zw}=\overline{z}\,\overline{w}$.
Hint. Multiply the two numbers out first and conjugate the result; then conjugate each factor and multiply. Both are two lines of real arithmetic.

Exercise 0.8 (extension) Find all four complex numbers z with $z^4=-4$.
Hint. Write -4 in polar form as $4\,e^{i\pi}$; a fourth root has modulus $4^{1/4}=\sqrt{2}$ and argument $\frac{1}{4}(\pi+2\pi k)$ for $k=0,1,2,3$.

Answer. $z\in\{1+i,\ -1+i,\ -1-i,\ 1-i\}$, i.e. $z=\pm1\pm i$.

Matrix algebra

Exercise 0.9 (diagnostic) Let $A=\begin{pmatrix}1 & 2\\ 0 & -1\end{pmatrix}$ and $B=\begin{pmatrix}3 & 1\\ 1 & 2\end{pmatrix}$. Compute AB, BA, and A^{T}, and verify $\operatorname{tr}(AB)=\operatorname{tr}(BA)$ even though $AB\neq BA$.
Answer. $AB=\begin{pmatrix}5 & 5\\ -1 & -2\end{pmatrix}$, $BA=\begin{pmatrix}3 & 5\\ 1 & 0\end{pmatrix}$, $A^{\mathsf{T}}=\begin{pmatrix}1 & 0\\ 2 & -1\end{pmatrix}$; $\operatorname{tr}(AB)=3=\operatorname{tr}(BA)$.

Exercise 0.10 (computational) Let $A=\begin{pmatrix}2 & 0\\ 1 & 3\end{pmatrix}$ and $B=\begin{pmatrix}1 & -1\\ 2 & 1\end{pmatrix}$. Compute AB and BA, and verify $\operatorname{tr}(AB)=\operatorname{tr}(BA)$ although $AB\neq BA$.
Answer. $AB=\begin{pmatrix}2 & -2\\ 7 & 2\end{pmatrix}$, $BA=\begin{pmatrix}1 & -3\\ 5 & 3\end{pmatrix}$; $\operatorname{tr}(AB)=4=\operatorname{tr}(BA)$.

Exercise 0.11 (computational) For $A=\begin{pmatrix}2 & 1\\ 0 & 2\end{pmatrix}$, compute A^2 and A^3.

Answer. $A^2=\begin{pmatrix}4 & 4\\ 0 & 4\end{pmatrix}$, $A^3=\begin{pmatrix}8 & 12\\ 0 & 8\end{pmatrix}$.

Exercise 0.12 (bridge) Let $A=\begin{pmatrix}1+i & 2-i\\ 0 & 3i\end{pmatrix}$ over $\mathbb{C}$. Write down the transpose A^{T} and the conjugate transpose $A^{\dagger}=\overline{A}^{\mathsf{T}}$.
Answer. $A^{\mathsf{T}}=\begin{pmatrix}1+i & 0\\ 2-i & 3i\end{pmatrix}$, $A^{\dagger}=\begin{pmatrix}1-i & 0\\ 2+i & -3i\end{pmatrix}$.

Exercise 0.13 (extension) For square matrices A,B of the same size, prove the cyclic identity $\operatorname{tr}(AB)=\operatorname{tr}(BA)$.
Hint. Write both traces as double sums over entries and note that the middle step only reorders a finite sum of scalars.

Exercise 0.14 (bridge) For matrices $A \in M_{m\times n}(\mathbb{F})$ and $B \in M_{n\times p}(\mathbb{F})$, prove that $(AB)^\mathsf{T} = B^\mathsf{T} A^\mathsf{T}$.
Hint. Compare the (i, j) entry of each side. The transpose turns $(AB)_{ji}$ into a sum in which the factors appear in the opposite order.

Determinants

Exercise 0.15 (diagnostic) Evaluate $\det\begin{pmatrix}1 & 2 & 3\\ 0 & 1 & 4\\ 5 & 6 & 0\end{pmatrix}$ and state whether the matrix is invertible.

Answer. $\det = 1$; invertible.

Exercise 0.16 (computational) Use the 2×2 formula to invert $A = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$.

Answer. $A^{-1} = \begin{pmatrix}-2 & 1\\ \frac{3}{2} & -\frac{1}{2}\end{pmatrix}$.

Exercise 0.17 (bridge) For $A = \begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}$, find all λ for which $\det(\lambda I - A) = 0$. (These values reappear in Week 5 as the eigenvalues of A.)
Answer. $\lambda = 1$ and $\lambda = 3$.

Exercise 0.18 (computational) Evaluate $\det\begin{pmatrix}1 & 2 & 1\\ 0 & 3 & 2\\ 1 & 0 & 1\end{pmatrix}$ and state whether the matrix is invertible.

Answer. $\det = 4$; invertible.

Exercise 0.19 (computational) Let $A = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B = \begin{pmatrix}2 & 0\\ 1 & 3\end{pmatrix}$. Compute $\det A$, $\det B$, and $\det(AB)$, and confirm $\det(AB) = \det A\, \det B$.
Answer. $\det A = -2$, $\det B = 6$, $\det(AB) = -12 = (-2)(6)$.

Exercise 0.20 (extension) Evaluate the block-diagonal determinant $\det\begin{pmatrix}1 & 1 & 0 & 0\\ 1 & 2 & 0 & 0\\ 0 & 0 & 2 & 0\\ 0 & 0 & 1 & 3\end{pmatrix}$.

Hint. A determinant with a zero off-diagonal block factors as the product of the determinants of the diagonal blocks.

Answer. $\det = 6$.

Exercise 0.21 (extension) For 2×2 matrices A, B, prove that $\det(AB) = \det A\, \det B$ directly from the entries.
Hint. Multiply AB out, expand $\det(AB)$, and watch four of the terms cancel in pairs; what is left factors as $(ad - bc)(eh - fg)$.

Gaussian elimination

Exercise 0.22 (diagnostic) Solve, by row reduction,

$$\begin{aligned} x + y + z &= 2, \\ x - y + 2z &= 5, \\ 2x + y - z &= 2. \end{aligned}$$

Answer. $(x, y, z) = (2, -1, 1)$ (unique).

Exercise 0.23 (diagnostic) Find a basis for the solution space (null space) of $Ax = 0$, where $A = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 3 & 1 \end{pmatrix}$, and state its dimension.

Hint. Row-reduce A, identify the pivot and free columns, then solve for the pivot variables in terms of the free ones.

Answer. Basis $\{(1, 1, 0, 0),\ (-2, 0, 1, 1)\}$; dimension 2.

Exercise 0.24 (computational) Compute A^{-1} by row-reducing the augmented block $[A \mid I]$, for $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$.

Answer. $A^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$.

Exercise 0.25 (computational) Solve, by row reduction,

$$\begin{aligned} x + y + z &= 6, \\ x - y + z &= 2, \\ 2x + y - z &= 1. \end{aligned}$$

Answer. $(x, y, z) = (1, 2, 3)$ (unique).

Exercise 0.26 (computational) Describe the full solution set of

$$\begin{aligned} x + 2y - z &= 3, \\ 2x + y + z &= 3, \end{aligned}$$

in the form "particular solution + span of the null space".

Answer. $(x, y, z) = (1, 1, 0) + t(-1, 1, 1)$, $t \in \mathbb{R}$.

Exercise 0.27 (computational) Compute A^{-1} by row-reducing the augmented block $[A \mid I]$, for $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Answer. $A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$.

Exercise 0.28 (bridge) Let $A \in M_{m\times n}(\mathbb{F})$. Prove that the null space $\ker A = \{x \in \mathbb{F}^n : Ax = 0\}$ is closed under addition and scalar multiplication (so it is a subspace of $\mathbb{F}^n$).

Hint. Matrix multiplication is linear in x, so $A(x + y)$ and $A(cx)$ split at once. Check $0 \in \ker A$ first.

Exercise 0.29 (extension) For which real values of a does the system

$$\begin{aligned} ax + y + z &= 1, \\ x + ay + z &= 1, \\ x + y + az &= 1 \end{aligned}$$

have a unique solution, infinitely many solutions, or no solution? Give the solution in the unique case.

Hint. Add the three equations to get $(a+2)(x+y+z) = 3$; subtract pairs to get $(a-1)(x-y) = 0$ and $(a-1)(y-z) = 0$. The determinant of the coefficient matrix is $(a-1)^2(a+2)$.

Answer. Unique iff $a \neq 1$ and $a \neq -2$, with $x = y = z = \dfrac{1}{a+2}$; infinitely many if $a = 1$ (the plane $x + y + z = 1$); no solution if $a = -2$.

Exercise 0.30 (diagnostic) Classify each displayed RREF augmented matrix as having no solution, a unique solution, or infinitely many solutions. For each consistent system, give its full solution set.

$$\text{(a)} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right], \qquad \text{(b)} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array}\right], \qquad \text{(c)} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & -1 \end{array}\right].$$

Answer. (a) Unique: $(2,-1)$. (b) No solution. (c) Infinitely many: $(x, y, z) = (4,-1,0) + t(2,-3,1)$, $t \in \mathbb{R}$.

Exercise 0.31 (repair) Row-reduce and classify each system, giving the full solution set when it is consistent:

$$\begin{array}{lll} \text{(a)} & x + y = 1, & x - y = 3, \\ \text{(b)} & x + y = 1, & 2x + 2y = 3, \\ \text{(c)} & x + y = 1, & 2x + 2y = 2. \end{array}$$

Answer. (a) Unique: $(2,-1)$. (b) No solution. (c) Infinitely many: $(x, y) = (1,0) + t(-1,1)$, $t \in \mathbb{R}$.

Separate analysis-readiness route

Take these five checks without notes in 15–20 minutes before the application weeks. They do not contribute to the algebra-readiness score.

Exercise 0.32 (analysis route) Differentiate $f(t) = t^2 e^{-t}$.

Answer. $f'(t) = e^{-t}(2t - t^2)$.

Exercise 0.33 (analysis route) Use integration by parts to evaluate $\displaystyle\int_0^1 te^t\,dt$.

Answer. 1.

Exercise 0.34 (analysis route) Find the general real solution of $y'' - 3y' + 2y = 0$.

Answer. $y(t) = C_1 e^t + C_2 e^{2t}$.

Exercise 0.35 (analysis route) Determine whether $\displaystyle\sum_{n=0}^{\infty} 3^{-n}$ converges and, if it does, find its sum.

Answer. It converges to $\frac{3}{2}$.

Exercise 0.36 (analysis route) For $f(x) = x$ on $[-\pi, \pi]$, compute the first Fourier sine coefficient $b_1 = \frac{1}{\pi}\int_{-\pi}^{\pi} x \sin x \, dx$.
Answer. $b_1 = 2$.

Reading your results

Read the three readiness bands above in order and stop at the first that applies, so a strand-based condition always outranks a score. Then follow only the route for a missed diagnostic strand. Work with the refresher open, close it for the named recheck, and require a correct result with working before declaring the strand repaired. Each recheck problem is held back from its own practice list, so it tests a repaired method on an input you have not already worked with notes open.

- **Complex arithmetic** (diagnostic 0.1–0.2): revisit *Refresher* §1; practise 0.3 and 0.6; recheck with 0.4 and 0.5. Problem 0.7 is a proof bridge and 0.8 is extension, neither scored for readiness.
- **Matrix algebra** (diagnostic 0.9): revisit *Refresher* §2; practise 0.10; recheck with 0.11, and for the same A also write down A^{T} and $\operatorname{tr} A$, which 0.11 alone does not test. Problem 0.12 is an adjoint bridge and 0.13–0.14 are proof extensions/bridges, excluded from readiness.
- **Determinants** (diagnostic 0.15): revisit *Refresher* §3; practise 0.16 and 0.19; recheck with 0.18. Problem 0.17 is the Week 5 characteristic-polynomial bridge; 0.20–0.21 are extensions.
- **Gaussian elimination** (diagnostic 0.22–0.23 and 0.30): revisit *Refresher* §4; practise 0.24–0.27; recheck consistency and full solution sets with 0.31, and give a basis for the null space of the coefficient matrix in its case (c), which 0.31 alone does not test. Problem 0.28 is a Lecture 1 subspace bridge and 0.29 is a parameter-family extension, neither scored for readiness.

Analysis routing. For problems 0.32–0.36, 4–5 correct means the later application bridge is ready; 2–3 means repair only the named topics; and 0–1 means arrange a calculus/methods review before those weeks. Route 0.32 to product/chain rules, 0.33 to integration by parts, 0.34 to constant-coefficient ODEs (before Lectures 1 and 6), 0.35 to geometric series (before Lectures 6 and 13), and 0.36 to Fourier coefficients (before Lecture 13). Use prior calculus/methods notes or tutorial support, work two similar examples, then redo the missed check without notes on a later day.